# 1 Estimating the Sum of a Series

Suppose we have been able to use the Integral Test to show that the series $\sum_{i=1}^{\infty} a_i$ converges. Since $\lim_{n\to\infty} s_n = s$, any partial sum $s_n$ is an approximation to $s$. We wish to estimate the size of the remainder

$$R_n := s - s_n = \sum_{i=n+1}^{\infty} a_i.$$

By the same reasoning as before, we prove the following theorem:

**Theorem 1.1** (Remainder Estimate for the Integral Test)**.** *Let $f$ be a continuous, positive, decreasing function for $x \geq n$. Suppose that $\sum_{n=1}^{\infty} a_n$ is convergent, where $a_n = f(n)$. Then*

$$\int_{n+1}^{\infty} f(x)dx \leq R_n \leq \int_{n}^{\infty} f(x)dx. \tag{1.2}$$

*Proof.* Just note that $R_n = \sum_{i=n+1}^{\infty} a_i$ is a right-Riemann sum for $\int_{n+1}^{\infty} f(x)dx$ and hence an overestimate (since $f$ is decreasing), while it is a left-Riemann sum for $\int_{n}^{\infty} f(x)dx$, and hence an underestimate. □

1. Use Equation (1.2) to show that

$$s_n + \int_{n+1}^{\infty} \leq s \leq s_n + \int_{n}^{\infty} f(x)dx \tag{1.3}$$

2. Consider the series $s = \sum_{n=1}^{\infty} \frac{1}{n^3}$.

   (a) Approximate $s$ by $s_{10}$. Find the maximum error in this approximation.

   (b) How many terms are required to ensure that the sum is accurate to within 0.0005?

   (c) Use Equation (1.3) with $n = 10$ to find an interval which contains the sum $s$.

   (d) Find the maximum error if one approximates $s$ by the midpoint of the interval in part (c). Compare this approximation to $s_{10}$.

3. Estimate the sum $\sum_{n=1}^{\infty}(2n+1)^{-6}$ correct to five decimal places.